Software Design Specifications

UNIVERSITY OF SARGODHA

DEPARTMENT OF COMPUTER SCIENCE

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Area: AI & ML

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Physics Informed LLM Based Weather Prediction System

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Definition of Terms, Acronyms and Abbreviations

This section provides the definitions of all terms, acronyms, and abbreviations required to interpret the terms used in the document properly.

Term	Description
NWP	Numerical Weather Prediction
PINNS	Physics Informed Neural Network System
LLMs	Large Language Model
ML	Machine Learning

1. Introduction

1.1. Purpose of Document

The purpose is of this document to present the design requirements of this project “Physics Informed LLM Based Weather Prediction System”. It will provide a detailed explanation of system’s component design, data flow, architecture and implementation strategies. To ensure an alignment between project requirements and design decisions for stakeholders, this document will assist as a reference and technical blueprint for developers. The intendent audience (project advisors, the project manager, the development team members, and the university evaluators) who will evaluate the system’s planed architecture and feasibility.

1.2. Project Overview

The main concern of this project is to create a hybrid weather prediction system that integrates Physics-Informed Neural Network (PINNS) with Machine Learning (ML) techniques and Large Language Models (LLMs). Traditional Numerical Weather Prediction (NWP) models are accurate and precise but require major computational power and time which may often lead to delays in real-time forecasting. On the other-hand, pure data driven ML models can deliver faster results but may generate outputs that breach physical laws for an instant unrealistic temperature point or wind pattern. In contrast, our system will aim to address these issues by integrating physical equations (e.g., Navier-Stokes for fluid dynamics) directly into the neural network training process through PINNs, allowing predictions are both data informed and physically consistent. LLMs will then process these outputs to produce natural language summaries, allowing the forecasts accessible to non experts. The main concern is to improve accuracy for short term predictions, less computational cost a compared to supercomputer based NWP and enhanced interpretability which can help in applications such as agriculture, disaster management, and policy planning.

1.3. Scope

The system will handle short-term weather forecasting (1-3 days ahead) for key variables like temperature, humidity, pressure, and wind speed, using publicly available datasets. It includes data preprocessing, PINN-based prediction with physics constraints, LLM integration for human-readable outputs, and basic evaluation against ML baselines. The system will be a prototype focused on demonstration, not production deployment. What the system will do: Process weather data, generate physics-consistent forecasts, and provide interpretable summaries. What the system will not do: Perform long-term climate modeling, high-resolution global simulations, real-time data ingestion from sensors, or serve as a public-facing operational tool. Exclusions also include training custom LLMs from scratch or using proprietary datasets.

The system will generate short term weather forecasting (1-3 days ahead) for key variables like temperature, pressure, humidity and wind speed using publicly available datasets.

2. Design Considerations

2.1. Assumptions

- Datasets like ERA5 are freely accessible.
- Minimum hardware requirements for Python/ML libraries.
- Pre-trained LLMs offer reliable explanations.

2.2. Dependencies

- External Libraries e.g., PyTorch, Hugging Face Transformers etc.
- Availability of team resource.

2.3. Risks and Volatile Areas

2.3.1. Data Quality and Availability Risks

Risk: There is a risk of dataset going temporarily unavailable and may have missing values.

Design Response:

- Implement robust data validation and cleaning pipelines
- Include missing value imputation strategies
- Provide fallback to alternative datasets if primary source fails

2.3.2. Physics Constraint Implementation Complexity

Risk: Encoding physics law (Navier-Stokes, thermodynamic equations) into neural network is a very complex task.

Design Response:

- Start with simplified physics constraints and incrementally add complexity.
- Use established libraries like DeepXDE that provide physics-informed loss implementations.
- Implement modular physics constraint functions that can be individually tested.
- Allow configurable weighting between data loss and physics loss.

2.3.3. Model Performance and Accuracy

Risk: Achieving target accuracy can be a hypothetical thing or may perform worse than baseline ML models

Design Response:

- Implement comprehensive evaluation framework with multiple metrics.
- Design modular architecture allowing easy swapping of model components.
- Include extensive hyperparameter tuning capabilities.
- Compare against multiple baselines (LSTM, CNN, pure ML models).

2.3.4. Computational Resource Limitations

Risk: Training PINNs with physics constraints may require more computational resources than available on moderate hardware.

Design Response:

- Implement efficient data batching and sampling strategies
- Use mixed-precision training to reduce memory footprint
- Design scalable architecture that can run on CPU with degraded performance Provide checkpointing to resume training after interruptions
- Optimize grid resolution and temporal sampling

3. References

Ref. No.	Document Title	Date	Source
[1]	Proposal_G38-SS1	2025	https://github.com/Abubakr4t/FYP-/blob/main/Proposal_G38-SS1.pdf
[2]	SRS-1.3	2025	https://github.com/Abubakr4t/FYP-/tree/main/SRS

4. Appendix

Workflow: Weather data → PINN forecast → LLM explanation → User output.